CSA-0619 CO1 AT1 ANSWERS

Q26. Application: Smart Grid Energy Distribution System An energy distribution system uses the recurrence $T(n)=4T(n/2)+n\log n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze its performance for large energy networks. Interpret efficiency using asymptotic notation.

Q26. Smart Grid Energy Distribution System — Master Theorem Analysis

1. Recurrence Relation

$T(n) = 4T(n/2) + n \log n$

2. Identifying Master Theorem Parameters

The general form of the Master Theorem is:

$T(n) = aT(n/b) + f(n)$

Comparing with the given recurrence:

Parameter	Value	Meaning
a	4	Number of subproblems (sub-grids/sub-networks)
b	2	Factor by which problem size is divided
f(n)	n log n	Cost of dividing + combining work at each level

3. Compute the Watershed Function $n^{\log_b a}$

$\log_b a = \log_2 4 = 2$
$n^{\log_b a} = n^2$

4. Compare f(n) with $n^{\log_b a}$

$f(n) = n \log n$
$n^{\log_b a} = n^2$

We need to check whether n log n is **polynomially smaller** than n^2.

Since:

$n \log n = O(n^{(2-\varepsilon)})$ for any ε, $0 < \varepsilon \le 1$ (e.g., $\varepsilon = 1$)

(because log n grows slower than any positive power of n)

This satisfies the condition for **Case 1** of the Master Theorem.

5. Case Selection — Justification

Master Theorem Case 1:
If $f(n) = O(n^{(\log_b a - \varepsilon)})$ for some $\varepsilon > 0$, then:

$T(n) = \Theta(n^{(\log_b a)})$

Here:

- $n^{(\log_b a)} = n^2$
- $f(n) = n \log n$ grows strictly slower than n^2 (polynomially smaller)
- Hence Case 1 applies directly.

6. Final Time Complexity

$T(n) = \Theta(n^2)$

7. Performance Analysis for Large Energy Networks

- The recurrence models a system where each node/region of the smart grid is recursively divided into **4 sub-networks** ($a = 4$), each of **half the size** ($b = 2$), with an additional **n log n** overhead per level for tasks such as load balancing, routing computation, or data aggregation.
- Since the divide step ($a=4$, $b=2$) dominates asymptotically over the combine step ($n \log n$), the overall complexity is governed by the recursive branching term $n^{(\log_b a)} = n^2$.
- For large-scale smart grids (large n = number of nodes/sensors/substations), the algorithm's running time grows **quadratically**, meaning doubling the network size roughly **quadruples** the processing time.
- This is less efficient than linear or n log n algorithms, but still polynomial and tractable for moderately large networks compared to exponential-time alternatives.

8. Interpretation Using Asymptotic Notation

Upper Bound (Big-O): $T(n) = O(n^2)$
Lower Bound (Big-Ω): $T(n) = \Omega(n^2)$
Tight Bound (Big-Θ): $T(n) = \Theta(n^2)$

Interpretation: The smart grid energy distribution algorithm exhibits **quadratic time complexity $\Theta(n^2)$**. This means computational cost scales as the square of the number of grid elements (nodes, sensors, or distribution points). It is efficient enough for medium-to-large networks but may require optimization (e.g., reducing branching factor a, or using divide-and-conquer with pruning) for very large-scale national or regional smart grid deployments.